

CHEATSHEET · QUICK REFERENCE

Lab 18 — Data at Rest (LUKS) and Data in Transit (TLS / SSH)

Encrypt a volume with LUKS and prove the ciphertext, serve TLS and inspect the negotiated cipher, harden SSH, and map retention to data classification.

LUKS at rest

```
cryptsetup luksFormat /dev/loop20 --key-file=<(echo -n 'LabPass!1')
Format an encrypted volume

cryptsetup luksOpen /dev/loop20 secret --key-file=<(echo -n 'LabPass!1')
Open the mapper device

cryptsetup luksDump /dev/loop20 | head -30
Inspect the LUKS header

strings /srv/secret.img | grep financials || echo no plaintext
Prove ciphertext is unreadable

cryptsetup luksClose secret
Close the volume
```

TLS in transit

```
openssl req -x509 -newkey rsa:2048 -nodes -keyout lab.key -out lab.crt -subj "/CN=lab.local" -days
Self-signed cert

ssl_protocols TLSv1.2 TLSv1.3;
Minimum protocols in nginx

echo | openssl s_client -connect 127.0.0.1:443 2>/dev/null | grep -E "Protocol|Cipher"
Inspect the negotiated cipher
```

SSH hardening (sshd_config)

```
PasswordAuthentication no
Keys only

KexAlgorithms curve25519-sha256
Modern key exchange

Ciphers chacha20-poly1305@openssh.com,aes256-gcm@openssh.com
AEAD ciphers only

grub-mkpasswd-pbkdf2
GRUB bootloader password hash
```

Retention by classification

Class	Encryption	Retention
Public	Optional	As needed
Internal	TLS in transit	3 years
Confidential	LUKS + TLS	7 years
Restricted	HW-crypt + TLS1.3 + MFA	7+ yrs, legal hold